

$$C_{\phi G}^{(11)} \rightarrow \frac{1}{16\pi^2}$$

$$\begin{aligned} & \left(-\frac{1}{24} \sum_{\mathbf{p}} g_1^2 g_3^2 \text{LF}_{3,0}[\mathbf{m}_{\tilde{d}}^{\mathbf{p}}] + \frac{1}{24} \sum_{\mathbf{p}} g_1^2 g_3^2 \text{LF}_{4,-1}[\mathbf{m}_{\tilde{d}}^{\mathbf{p}}] - \frac{1}{24} g_3^2 (6 \overline{y_u^{\mathbf{p}r}} y_u^{\mathbf{p}r} + \sum_{\mathbf{p}} g_1^2) \text{LF}_{3,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \right. \\ & \frac{1}{24} g_3^2 (6 \overline{y_u^{\mathbf{p}r}} y_u^{\mathbf{p}r} + \sum_{\mathbf{p}} g_1^2) \text{LF}_{4,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \frac{1}{12} \sum_{\mathbf{p}} g_1^2 g_3^2 \text{LF}_{3,0}[\mathbf{m}_{\tilde{u}}^{\mathbf{p}}] - \\ & \frac{1}{12} \sum_{\mathbf{p}} g_1^2 g_3^2 \text{LF}_{4,-1}[\mathbf{m}_{\tilde{u}}^{\mathbf{p}}] - \frac{1}{4} g_3^2 \overline{y_u^{\mathbf{p}r}} y_u^{\mathbf{p}r} \text{LF}_{3,0}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}] + \frac{1}{4} g_3^2 \overline{y_u^{\mathbf{p}r}} y_u^{\mathbf{p}r} \text{LF}_{4,-1}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}] - \\ & \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{2,2,0}[\mathbf{m}_{\tilde{d}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] - \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{d}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \\ & \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{d}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{d}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] - \\ & \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{d}}^{\mathbf{r}}] + \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{d}}^{\mathbf{r}}] + \\ & \frac{1}{4} g_3^2 \tilde{\mu}^2 \overline{y_d^{\mathbf{p}r}} y_d^{\mathbf{p}r} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{d}}^{\mathbf{r}}] - \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{2,2,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] - \\ & \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] + \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] + \\ & \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{q}}^{\mathbf{p}}, \mathbf{m}_{\tilde{u}}^{\mathbf{r}}] - \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \\ & \left. \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] + \frac{1}{4} g_3^2 \overline{a_u^{\mathbf{p}r}} a_u^{\mathbf{p}r} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{u}}^{\mathbf{r}}, \mathbf{m}_{\tilde{q}}^{\mathbf{p}}] \right) \end{aligned}$$